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Applied Information Extraction in Python

Today's goal • 3/3 items

Module 1
Information Extraction

Module 2
Named Entity Recognition (NER)

Named Entities and Named Entity Recognition

- What is Named Entity Recognition (NER)? Video • 5 min
- Knowledge Check: Named Entities and Named Entity Recognition Practice Assignment • Grade: 100%

Setting up NER as a Machine Learning Task

- BIO Encoding for Named Entity Labels Reading • 10 min
- BILOU Encoding for Named Entity Labels Reading • 10 min
- Machine Learning Fundamentals: How Machines Learn to Label Named Entities Reading • 10 min
- NER as a Sequence Classification Task Video • 8 min
- Knowledge Check: Setting up NER as a Machine Learning Task Practice Assignment • Grade: 100%

Hidden Markov Models (HMMs)

- Markov Chain and Hidden Markov Models Reading • 10 min
- Fundamentals of Markov Chain Models Video • 7 min
- Hidden Markov Models (HMMs) Video • 11 min
- Training Hidden Markov Models: How HMMs Learn to Assign Labels Reading • 10 min
- **The Math Behind HMMs: How Probabilities Power Sequence Labeling** Reading • 10 min
- Conditional Random Fields (CRFs) Video • 6 min
- Knowledge Check: Hidden Markov Models (HMMs) Practice Assignment • 15 min

How to Extract Information Using CRFs in Python?

- Demo: CRF Model Training Video • 8 min
- Jupyter Notebook Practice on Training CRFs Lab • 1h

Module Wrap-Up & Assignment

- Assignment 2 Introduction: Implementing a CRF Model Video • 4 min
- Build an Information Extraction Pipeline for CRF Based Extraction Programming Assignment • 3h
- Module 2 Assignment Graded Assignment • 30 min

Module 3
Neural Network Models

Module 4
Transformers Transform Information Extraction

The Math Behind HMMs: How Probabilities Power Sequence Labeling

0:18 / 4:16 2x

Why Do We Need Math in HMMs?

You now know what a Hidden Markov Model (HMM) is and how it's trained. But underneath the transitions and emissions lies a layer of math that gives HMMs their power.

Don't worry, we won't drown in equations. This reading will give you just enough mathematical understanding to follow what's happening, evaluate results, and get ready for more advanced models like Conditional Random Fields (CRFs).

HMMs are **probabilistic models**, which means every prediction is based on likelihoods, how probable a label is given what we've seen.

Math helps us:

- Determine how likely a sequence of words is
- Decide the most probable path of hidden states (labels)
- Evaluate which parameters or model settings are better

Without math, HMMs are just labeled boxes. With math, they become powerful predictive tools.

The Three Core Problems HMMs Are Built to Solve

Every HMM tackles three classic questions:

1. **Evaluation** Given an HMM and a sequence of words, how likely is that sequence? This helps us compare models or interpret confidence scores.
2. **Decoding** Given a sequence of words, what is the most likely sequence of labels (states)? This is often solved using the **Viterbi algorithm**.
3. **Learning (Training)** Given only the words (and maybe partial labels), how do we estimate the model parameters? You saw this earlier with the **Baum-Welch algorithm**, a form of Expectation-Maximization (EM).

The Forward Algorithm (Simplified)

Let's zoom in on the **Evaluation** problem: How do we compute the total probability of a sentence?

That's where the **Forward Algorithm** comes in.

Basic idea:

- At each time step (word in the sentence), the model computes all possible paths leading up to that word
- It does this recursively, combining prior probabilities with transition and emission probabilities
- This gives you the **total probability** that the model would generate the observed sentence

Example:

"Dr. Jones works at Red Cross."

We want to know:

What's the probability this sentence was generated by our HMM, considering all possible label sequences?

Instead of brute-forcing all options, the Forward Algorithm efficiently sums the probabilities by:

- Starting from **initial probabilities**
- Iteratively combining **transitions** and **emissions**
- Ending with a final **total probability score**

This process is both efficient and elegant, and it's fundamental to how HMMs evaluate inputs.

What Are We Really Calculating?

Let's demystify this a bit.

Suppose:

- You're at **time step t**
- You're considering **state j** (e.g., PER)
- You've already calculated probabilities for all states at **time t–1**

Then the forward step involves:

- The **transition probability** from each previous state to state j
- The **emission probability** of the current word from state j
- The **accumulated forward probability** up to that point

The model multiplies and sums these values to compute the likelihood of being in state j at time t, given everything so far.

Why Does This Matter in NLP?

You're about to work with real data. That means:

- **Scores will matter.** The math tells you how confident the model is.
- You'll use **decoding** (e.g., Viterbi) to find the best sequence of labels.
- Understanding the **structure behind prediction** helps with debugging, tuning, and justifying model decisions.

In short: if you can't follow the math, you can't fully trust what your model is doing.

Recommended Resources

- Speech and Language Processing (Jurafsky & Martin) – Chapter on HMMs
  - Chapter 17: Sequence Labeling for Parts of Speech and Named Entities)
- Blog: "HMMs for Sequence Tagging" – by Michael Collins Clear walkthrough of HMMs in the context of NLP
- HMM– Baum Welsh and Viterbi Algorithms (YouTube)

Reflection Prompt

You've learned that HMMs use the Forward Algorithm to evaluate sentence probability.

Imagine two models, one trained on **medical notes**, and the other on **social media posts**. The same sentence gets **two different likelihood scores**.

What could that tell you about the models? What might each be "expecting" to see?

Assignment Preview

As you explore how CRFs improve upon HMMs, keep this math in mind, especially how probabilities shape prediction.

You're not just tagging words, but reasoning over sequences.

Go to next item ✓ Completed

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